Can a nation-state open your locked files?

Scenario: attacker has the computer, the encrypted files, the keybulk file and the full source code. Password: 24 characters - 10 numbers, 10 mixed-case letters, 4 special characters.

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BRUTAL HONEST VERDICT

They will **not** break the encryption by guessing. If they get in, it will be by a side door: the computer's memory, leftover copies of your files, malware, or pressure on you. The lock is excellent; the house around it decides the outcome.

1. Guessing the password: practically impossible

If the password is **randomly generated** (the app's Generate button), there are about 2^{140} possible passwords. Each guess must run Argon2id, which deliberately takes ~0.2 s and 128 MB of memory. Even granting an impossible **one trillion guesses per second**, finding it takes about **28,000,000,000,000,000,000,000 years** - roughly two trillion times the age of the universe. Quantum computers do not change this: they can at best halve the 'difficulty exponent', which still leaves an impossible job. Having the source code does not help them - good encryption is designed to stay secure even when everything except the password is public.

The catch: if *you* invent the password from names, dates, phone numbers or keyboard patterns (e.g. Rahul@1998India...), attackers try those first. Real strength can drop from 140 bits to 40-50 bits: **days to months** for a state with a GPU farm.

2. The side doors (where a real attacker would go)

Computer seized while unlocked, asleep or hibernated	The master key sits in memory or hiberfil.sys and can be lifted in minutes.
Malware or keylogger installed before seizure	Your password is recorded as you type it. The math never gets tested.
SSD / pen-drive leftovers	Shredding can't reach every old flash block. Original files may be forensically recoverable.
Copies outside the locker	Thumbnails, Office autosave, temp files and cloud sync live on C:, which the app never touches.
Visible details	File names, sizes and dates of locked files are readable (e.g. Budget.xlsx.dlock).
Pressure on you	Legal compulsion, threats, or a password written down somewhere.
Young, unaudited code	Built and tested, but no outside security firm has reviewed it yet.

3. Scorecard

Encryption math vs brute force, random password, PC powered off	9.5
Same, but password made from personal info or patterns	4
PC grabbed while running, asleep or hibernated	2
Files that lived unencrypted on an SSD before locking	5
Overall, real world, against a nation-state	7

4. How to push it toward 9.5

• Use the **Generate** button; memorise it, never reuse it. • Click **Lock** and **shut down** - never just sleep; turn hibernation off (`powercfg /h off`). • Turn on **BitLocker** for C: and the data drives so leftovers are encrypted too. • Create sensitive files directly on an encrypted drive. • Keep antivirus on and the PC physically secure. • Get an independent code audit before trusting it with life-or-death data.